

1(a) Define X^2 variate? Derive the sampling distribution of Chi-square (X^2) variate?

Answer: X^2 - variate / X^2 -statistic: The square of a standard normal variate is known as a chi-square (X^2) variate with 1 d.f.

This if $X \sim N(\mu, \sigma^2)$ then $z = \frac{X-\mu}{\sigma} \sim N(0, 1)$ And $z^2 = \left(\frac{X-\mu}{\sigma}\right)^2$ is a x^2 - variate with 1 d.f.

In general, if X_i ($i=1,2,\dots,n$) are in independent normal variate with mean μ_i and

variance σ_i^2 ($i=1,2,\dots,n$), then $X^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2$ is a X^2 - variate / X^2 -statistic with n degree of freedom (d.f.).

Derivation of the X^2 - distribution: If X_i ($i = 1, 2, \dots, n$) are independent $N(\mu_i, \sigma_i^2)$

variate then the quality $X^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2$ is distribute as X^2 with n d.f.

Proof: Let, $z_i = \left(\frac{X_i - \mu_i}{\sigma_i}\right)$, $i = 1, 2, \dots, n$

Since $X_i \sim N(\mu_i, \sigma_i^2)$ then $Z_i \sim N(0, 1)$

Now the moment generating function of X^2 is given by

$$M_{X^2}(t) = M \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i}\right)^{2(t)} = M \sum_{i=1}^n Z_i^{2(t)} = \prod_{i=1}^n M_{Z_i^2}(t)$$

$$\text{But } M_{Z_i^2}(t) = E(e^{t Z_i^2}) = \int_{-\infty}^{\infty} e^{t Z_i^2} f(Z_i) dZ_i$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{t Z_i^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} Z_i^2}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{\infty} e^{-\left(\frac{1-2t}{2}\right) Z_i^2} dz_i$$

$$\Rightarrow \frac{2}{\sqrt{2\pi}} \cdot \frac{\left(\frac{1-2t}{2}\right)^{-\frac{1}{2}} \sqrt{\frac{1}{2}}}{2}$$

$$\Rightarrow \frac{1}{(1-2t)^{1/2}} \quad \text{Zetu, } \sqrt{\frac{1}{2}} = \sqrt{\pi}$$

$$M_{X^2}(t) = \sum_{i=1}^n M_{X^2}(t) = \sum_{i=1}^n \frac{1}{(1-2t)} \frac{1}{2} = \frac{1}{(1-2t)^{n/2}}$$

$$\Rightarrow \left(\frac{1/2}{1/2-t} \right)^{n/2}$$

Which is the moment generating function of a gamma variate with parameter

$$\lambda = \frac{1}{2} \quad \text{and} \quad r = \frac{n}{2} \quad [\text{since } G(\lambda, r) = \frac{\lambda^r}{\sqrt{r}} e^{-\lambda x} x^{r-1}]$$

Therefore the pdf of X^2 is

$$df(X^2) = \frac{\left(\frac{1}{2}\right)^{n/2}}{\sqrt{\frac{n}{2}}} e^{-\frac{1}{2}X^2} (X^2)^{\frac{n}{2}-1} dX^2 = \frac{e^{-\frac{X^2}{2}} (X^2)^{n/2-1}}{2^{n/2} \sqrt{n/2}} dX^2;$$

$$\text{or, } f(X^2) = \frac{\left(\frac{1}{2}\right)^{n/2}}{\sqrt{\frac{n}{2}}} e^{-\frac{1}{2}X^2} (X^2)^{\frac{n}{2}-1}$$

$$\Rightarrow \frac{e^{-\frac{X^2}{2}} (X^2)^{n/2-1}}{2^{n/2} \sqrt{n/2}} ; \quad 0 \leq X^2 \leq \infty$$

Which is the pdf of a X^2 - distribution with nd.f.

1(b) Show that X^2 -distribution approaches to normal distribution as $n \rightarrow \infty$.

OR, show that X^2 -distribution tends to normal distribution for large degree of freedom

Answer:-

$$\text{Let } z = \frac{x^2 - E(x^2)}{\sqrt{va_r(x^2)}} = \frac{x^2 - n}{\sqrt{2n}} = \frac{x^2}{\sqrt{2n}} - \sqrt{\frac{n}{2}}$$

The mgf of standard x^2 - variate z is given by

$$M_z(t) = E(e^{tz}) = E \left[e^{\left(\frac{x^2}{\sqrt{2n}} - \sqrt{\frac{n}{2}} \right) t} \right]$$

$$= e^{-\sqrt{\frac{n}{2}} t} E \left(e^{\frac{x^2 t}{\sqrt{2n}}} \right)$$

$$= e^{-t \sqrt{\frac{n}{2}}} M_{X^2} \left(\frac{x^2 t}{\sqrt{2n}} \right)$$

$$= e^{-t \sqrt{\frac{n}{2}}} \left(1 - \frac{2t}{\sqrt{2n}} \right)^{-\frac{n}{2}}$$

$$\Rightarrow M_z(t) = e^{-t \sqrt{\frac{n}{2}}} \left(1 - t \sqrt{\frac{2}{n}} \right)^{-\frac{n}{2}}$$

Taking logarithm of both sides then we get

$$\begin{aligned}
\log M_z(t) &= -t\sqrt{\frac{2}{n}} - \frac{n}{2} \log(1 - t\sqrt{\frac{2}{n}}) \\
&\Rightarrow -t\sqrt{\frac{n}{2}} + \frac{n}{2} \log(1 - t\sqrt{\frac{2}{n}}) \\
&= -t\sqrt{\frac{n}{2}} + \frac{n}{2} \left[t\sqrt{\frac{2}{n}} + \frac{2t^2}{2n} + \frac{t^3}{2} \left(\frac{2}{\sqrt{n}}\right)^3 + \dots \right] \\
&= -t\sqrt{\frac{n}{2}} + t\sqrt{\frac{n}{2}} + \frac{1}{2}t^2 + 0\left(\frac{1}{\sqrt{n}}\right) \\
&= \frac{t^2}{2} + \left(\frac{1}{\sqrt{n}}\right)
\end{aligned}$$

Where $0\left(\frac{1}{\sqrt{n}}\right)$ are terms containing $\sqrt{n}$ and higher power as n in the denominator.

$$\begin{aligned}
\text{So, } \prod_{n \rightarrow \infty} M_z(t) \frac{t^2}{2} + 0 &= \frac{1}{2}t^2 \\
&= \prod_{n \rightarrow \infty} M_z(t) = e^{\frac{1}{2}t^2}
\end{aligned}$$

Which is the mgf of a standard normal variate. Thus as $n \rightarrow \infty$ a standard X^2 -variate approach as standard normal variate. For large degree of freedom X^2 -distribution tends to normal distribution.

1(c) Write down the properties and applications of X^2 -dist².

Answer: Properties of X^2 -distribution : The properties of X^2 -distribution are given below:-

1)The pdf of X^2 -distribution with n d.f is $f(X^2) =$

$$\frac{e^{-\frac{X^2}{2}} (X^2)^{n/2-1}}{2^{n/2} \sqrt{n} \sqrt{2}} ; \quad 0 \leq X^2 \leq \infty$$

2)The range of X^2 -distribution is zero to infinity.

3)The mean of X^2 -distribution is equal to its d.f and the variance is equal to its twice d.f.

4)This distribution is positively skewed and leptokurtic.

5)Its mode is $n-2$.

6) X^2 distribution approaches normal distribution as $n \rightarrow \infty$

7)If $X \sim N(0, 1)$ then $X^2 \sim X^2$ with 1 d.f.

Application of X^2 distribution: X^2 distribution has a large no of application in statistics, some of which are enumerated below:-

1)To test if the hypothetical value of the population variance is $\sigma^2 = \sigma_0^2$ (raw)

2)To test the goodness of fit.

- 3) To test the homogeneity of independent estimates of the population variance.
- 4) To test the independence of attributes.
- 5) To test the homogeneity of independent estimates of the population coefficient.

2(a) Define a student's t-statistic and derive its probability density function.

Answer:-

student's t-statistic: A student's t-statistic, often simply referred to as the t-statistic, is a statistical measure that quantifies how far a sample mean is from the population mean in relation to the variability in the sample data. It is commonly used in hypothesis testing to assess whether a sample mean is significantly different from a hypothesised population mean when the population standard deviation is unknown or when dealing with small sample sizes. The t-statistic is calculated by dividing the difference between the sample mean and the hypothesised population mean by an estimate of the standard error of the sample mean. It follows a t-distribution, which takes into account the sample size and is used to determine the statistical significance of the observed difference.

probability density function:

Let $x_i = (i = 1, 2, \dots, n)$ be a random sample of size n from a normal population with mean μ and variance σ^2 then student's t is defined by the statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \text{ where } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \& \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Derivation of student's t-distribution:

Statement: If $x_1, x_2, \dots, x_n$ is a random sample from normal population with mean μ and variance σ^2 then student's t is defined by the statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \text{ where } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \& \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Proof: Let $x_1, x_2, \dots, x_n$ be a random from normal population with mean μ and variance σ^2 the sample $\bar{x}$ follows normal distribution with mean μ and variance $\frac{\sigma^2}{n}$.

Therefore $u = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ is a normal variate with mean zero and variance unity. The

$$\text{quality } v = \frac{(n-1)s^2}{\sigma^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2} \text{ follows } \chi^2\text{-distribution with } (n-1) \text{ d.f}$$

For normal population the sample mean $\bar{x}$ and the sample variance s^2 are independently distributed. In Other words , u and v are independently distributed .

Therefore the ratio $t = \frac{u}{\sqrt{\frac{v}{n-1}}} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$,

I.e the ratio between standardised normal variate and the root of mean x^2 follows t-distribution with (n-1) d.f. Therefore , $df(t) = \frac{dt}{\sqrt{n-1} \beta\left(\frac{1}{2}, \frac{n-1}{2}\right) \left(1 + \frac{t^2}{n-1}\right)^{n+1/2}}$; $-\infty < t < \infty$

Which follows the student's t-distribution with (n-1) d.f.

The probability density function (PDF) of the Student's t-distribution with $n-1$ freedom is given by:

$$f(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}$$

Where:

$f(t)$ is the PDF of the t-distribution. Γ represents the gamma function.

n is the degrees of freedom, which in this case is $n-1$ due to the sample size n .

This PDF describes the probability distribution of the t-statistic when you have a random sample from a normal population.

So, in summary, the t-statistic is used for comparing sample means to population means when dealing with small samples and unknown population standard deviations. Its PDF, as given above, characterises how t-values are distributed under these conditions.

2(c) Write down the properties and uses of f-distribution .

Answer:

The properties of f-distribution are given below-

1) The density function of f-distribution with n_1 & n_2 d.f in

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\left(\frac{n_1}{2}-1\right)}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}$$

2) The mean, variance and mode of this distribution are

$$\mu_1' = \frac{n_2}{n_2-2}, n_2 > 2; \mu_2 = \frac{(n_2)^2}{(n_2-2)^2} \left[\frac{2(n_1+n_2-2)}{n_1(n_2-4)} \right], n_2 > 4$$

$$\text{and } F = \frac{(n_1-2)n_2}{n_1(n_2+2)} \text{ Respectively.}$$

3) If $x \sim F(n_1, n_2)$ then $Y = \frac{1}{x} \sim F(n_2, n_1)$

4) The t^2 -distribution is the special case of the f-distribution where the numerator has only one d.f.

5) If denominator d.f n_2 tends to infinity then $n_1 F$ tends to χ^2 distribution with n_1 d.f.

6) As the d.f increases for both numerator and denominator the f-distribution tends to normal distribution.

Some uses are given below:

- To test the homogeneity of means in analysis of variance;
- To test the hypothesis on equality of two population variances to $\sigma_1^2 = \sigma_2^2$;
- To test the significance of correlation ratio;
- To test the significance of multiple correlation coefficient;
- To test non linearity of regression;
- To test the significance of inter class correlation coefficient.

3(a) Define point estimator. What are the desirable criteria of good estimator

Ans: A point estimator in statistics is a statistical method or formula that calculates a single, specific value (a point estimate) to estimate an unknown population parameter based on sample data.

A good estimator in statistics should possess several desirable criteria to provide accurate and reliable estimates of population parameters. These criteria include:

1. **Unbiasedness:** An estimator is unbiased if, on average, it produces estimates that are equal to the true population parameter. In other words, the expected value of the estimator is equal to the parameter it's estimating.

2. Efficiency: An efficient estimator has the smallest possible variance among all unbiased estimators. It provides estimates with the least amount of spread or variability around the true parameter value.
3. Consistency: A consistent estimator becomes increasingly accurate as the sample size grows larger. In practical terms, this means that as you collect more data, the estimator's estimates converge to the true parameter value.
4. Sufficiency: A sufficient estimator uses all relevant information contained in the sample data to estimate the population parameter. It doesn't leave out any important data.
5. Robustness: A robust estimator is not highly sensitive to the presence of outliers or deviations from the underlying assumptions. It should provide reasonably accurate estimates even when the data doesn't perfectly follow the assumed model.
6. Minimum Mean Squared Error (MSE): A good estimator minimises the mean squared error, which is the average of the squared differences between the estimates and the true parameter values. This balances bias and variance, providing a more accurate overall estimate.
7. Conservation of Information: An estimator should retain as much information from the sample as possible, ensuring that valuable data is not lost in the estimation process.
8. Ease of Computation: Practical estimators should be computationally feasible and not overly complex to calculate.
9. Asymptotic Normality: In large samples, a good estimator follows a normal distribution. This property is important for making inferences and constructing confidence intervals.

3(b) What are the different methods of point estimation? Describe the principles of maximum likelihood method of estimation.

Answer: Point estimation methods in statistics aim to estimate population parameters using sample data. One of the most widely used point estimation methods is the Maximum Likelihood Estimation (MLE). Here, I'll describe the principles of the Maximum Likelihood Method of estimation and provide an overview of other point estimation methods:

Maximum Likelihood Estimation (MLE):

Principles:

1. **Likelihood Function:** MLE begins with defining a likelihood function, denoted as $L(\theta; x)$, where θ represents the parameter to be estimated, and x is the sample data. The likelihood function measures the probability of observing the given data under different values of θ .
2. **Maximising Likelihood:** MLE seeks the value of θ that maximises the likelihood function. In other words, it finds the parameter value that makes the observed data most probable. This is often done by taking the derivative of the likelihood function with respect to θ and setting it equal to zero.
3. **Log-Likelihood:** For computational convenience, the likelihood function is often transformed into its natural logarithm, resulting in the log-likelihood function, denoted as $\ell(\theta; x)$. The log-likelihood simplifies calculations and retains the same maximum as the likelihood function.
4. **Consistency:** Under certain regularity conditions, MLE estimators are consistent, meaning that as the sample size increases, they converge to the true parameter value. This property ensures that with larger samples, MLE produces increasingly accurate estimates.
5. **Efficiency:** MLE estimators are often asymptotically efficient, meaning they achieve the smallest possible variance among unbiased estimators as the sample size increases.

Other Point Estimation Methods:

1. **Method of Moments (MoM):**
 - Match sample moments (e.g., mean, variance) with population moments to estimate parameters.
2. **Minimum Mean Squared Error (MMSE):**
 - Minimise the expected mean squared error between the estimate and the true parameter value.
3. **Bayesian Estimation:**
 - Use the Bayesian framework, combining a prior distribution with likelihood to obtain a posterior distribution from which point estimates (e.g., mean, median, mode) can be derived.
4. **Quantile Estimation:**

- Estimate specific percentiles of the population distribution directly from sample data.

5. Percentile Estimation:

- Estimate population percentiles (e.g., median, quartiles) using sample data.

In summary, MLE is a powerful point estimation method that seeks to maximise the likelihood of observing the data under different parameter values. It is widely used in statistics due to its desirable properties, including consistency and asymptotic efficiency. Other point estimation methods are also employed depending on the nature of the problem and the assumptions about the data and population distribution.

5(c) The average IQ of working women in Bangladesh is suspected to be more than 110, on average. A random sample of 64 of such women yielded an average IQ of 115.5 and a standard deviation of 20. Can you conclude that the average score of the women in the population is really more than 110? Test this at 5 percent level of significance.

Solution: The test is a one tailed one with

1. $H_0: \mu = 110$ against the alternative $H_1: \mu > 110$
2. $\alpha = 0.05$
3. Critical region: $z > 1.64$, where $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$
4. $\bar{x} = 115$, $n = 64$, $\mu_0 = 110$ and an estimate of σ is $s = 20$
5. $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{115.5 - 110}{20/\sqrt{64}} = 2.2$
6. Conclusion: The computed z value is significant at 5 percent level ($p < 0.05$). Hence we reject the null hypothesis and accept the alternative: the mean of working women is higher than the average (110) for women in the population in question. Confidence limits computation shows that the true mean lies between 111.4 and 119.6 .

Write down the properties and uses of t-distribution.

Answer: some properties of t-distribution are given below

1) The density function of t- distribution with n d.f is

$$f(t) = \frac{dt}{\sqrt{n-1} \beta\left(\frac{1}{2}, \frac{n-1}{2}\right) \left(1 + \frac{t^2}{n-1}\right)^{n+1/2}}; -\infty < t < \infty$$

2) It is an exactly sampling distribution derived from standard normal and χ^2 distribution derived from standard normal and χ^2 distribution;

3) All odd order moments are equal to zero;

4) For large d.f, t-distribution tends to standard normal distribution.

5) If $n=1$, t-distribution follows cauchy distribution

Some uses of t-distribution are given below:

a) It is used to test the population mean ($\bar{x}$) differs significantly from the hypothetical value of the population of the population mean.

b) It is used to test the significance of the difference between two population means.

c) It is used to test significance of an observed sample correlation coefficient and sample regression coefficient.

d) It is used to test the significance of observed partial and multiple correlation coefficients.

Show that for large d.f. T-distribution tends to standard normal distribution.

Proof: We know that pdf of t - distribution with n d.f is given by,

$$\begin{aligned} f(t) &= \frac{1}{\sqrt{n} \beta\left(\frac{n}{2}, \frac{1}{2}\right) \left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}; -\infty < t < \infty \\ &= \frac{\sqrt{\frac{n+1}{2}}}{\sqrt{n} \sqrt{\frac{n}{2}} \sqrt{\frac{1}{2}} \left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}} \\ &= \frac{\left(\frac{n-1}{2}\right)! \left(1 + \frac{t^2}{n}\right)^{-1/2}}{\sqrt{n\pi} \left(\frac{n-2}{2}\right)! \left(1 + \frac{t^2}{n}\right)^{n/2}} \end{aligned}$$

By stirling approximate formula for $n! \approx \sqrt{2\pi} e^{-n} n^{n+1/2}$ we have

$$f(t) = \frac{\sqrt{2\pi} e^{-\frac{n-1}{2}} \left(\frac{n-1}{2}\right)^{n/2} \left(1 + \frac{t^2}{n}\right)^{-1/2}}{\sqrt{n\pi} \sqrt{2\pi} e^{-\frac{n-2}{2}} \left(\frac{n-2}{2}\right)^{\left(\frac{n-1}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{\left(\frac{n}{2}\right)}}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{n\pi}} e^{-\frac{1}{2}} \left(\frac{n-1}{n-2} \right)^{\left(\frac{n}{2}\right)} \left(\frac{n-2}{2} \right)^{\left(\frac{1}{2}\right)} \left(\frac{\left(1+\frac{t^2}{n}\right)^{\left(-\frac{1}{2}\right)}}{\left(1+\frac{t^2}{n}\right)^{\left(-\frac{1}{2}\right)}} \right) \\
&= \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} \frac{\left(1-\frac{1}{n}\right)^{\left(\frac{n}{2}\right)}}{1-\frac{2}{n}} \left(1 - \frac{2}{n}\right)^{1/2} \left(\frac{\left(1+\frac{t^2}{n}\right)^{\left(-\frac{1}{2}\right)}}{\left(1+\frac{t^2}{n}\right)^{\left(-\frac{1}{2}\right)}} \right) \right)
\end{aligned}$$

$$\text{Now } \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} \frac{e^{-\frac{1}{2}}}{e^{-1}} \cdot 1 \cdot \frac{1}{e^{\frac{t^2}{2}}}$$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$$

Which is the standard normal distribution. Hence for large degrees of freedom, t-distribution tends to standard normal distribution.